CareSpeak: A Generative Voice-Based Training Tool for APRN Communication Skills

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Abstract

Advanced Practice Registered Nurse (APRN) students face limited opportunities to develop and refine communication skills for emotionally sensitive conversations, such as end-of-life discussions or counseling adolescents. Traditional standardized patient encounters are resource intensive and difficult to scale. We present CareSpeak, a proof-of-concept generative AI system designed to simulate real-time voice-based patient conversations. Developed with the input of clinical faculty, the system enables immersive training aligned with educational goals. CareSpeak integrates speech recognition, large language models, and text-to-speech to simulate emotionally responsive patient interactions, while faculty monitor live sessions and provide targeted feedback. This paper outlines the system architecture, compares related platforms, and discusses implementation outcomes and future directions.

CCS Concepts

• **Applied computing** → **Computer-assisted instruction**; *Health care information systems*; • **Computing methodologies** → *Natural language generation.*

Keywords

Healthcare simulation, nursing education, conversational AI, large language models, voice interface, communication skills, APRN training, generative AI

1 Introduction

Effective communication is fundamental to nursing care, especially in emotionally complex scenarios such as delivering a chronic disease diagnosis, discussing end-of-life options, or engaging patients from populations of gender and sexual minorities. However, many Advanced Practice Registered Nurse (APRN) students report insufficient training in such conversations due to placement constraints, lack of emotionally realistic simulations, and limitations of current teaching modalities. Standardized patient programs, while effective, are costly and unscalable. To address this, we developed CareSpeak, an AI-based conversational training tool designed to simulate high-empathy clinical scenarios using real-time voice interaction. Built with feedback from practicing clinicians, CareSpeak empowers students to practice skills requiring high emotional intelligence in a low-risk pedagogically structured environment.

2 Related Work

Simulation plays a crucial role in healthcare education, and there are several platforms to facilitate clinical skills training. Commercial offerings like Oxford Medical Simulation (OMS) provide high-fidelity virtual reality environments with more than 200 scenarios, including emotionally charged conversations, using voice recognition and AI-driven adaptive branching [5]. However, OMS relies primarily on scripted scenario logic rather than fully generative dialogue. Similarly, Elsevier's Shadow Health platform offers text-driven digital clinical experiences where nursing students interact with virtual patients using free text input matched with a large database of predefined responses, focusing on cultural and emotional competence [2]. Although valuable, both platforms are limited by their reliance on pre-defined or scripted interactions, which may not fully capture the spontaneity and variability of real patient conversations.

Concurrently, academic research is actively exploring the potential of large language models (LLMs) to create more dynamic simulated patients. Li et al. [4] demonstrated the feasibility of using LLMs to simulate patients for clinical interview practice. Furthermore, Holderried et al. [3] developed a simulated patient powered by GPT that incorporated automated feedback mechanisms, highlighting the potential of LLMs not only to simulate interactions but also to contribute to student evaluation. These studies underscore the promise of LLMs in educational applications, but often focus on specific tasks such as information gathering.

A significant challenge, particularly relevant for APRN training, is effectively simulating the emotional nuance required for sensitive conversations, such as end-of-life discussions or counseling diverse populations. Addressing the need for more sophisticated conversational agents, the GuideLLM framework proposed by Duan et al. [1] offers a structured approach for creating multimodal empathetic agents. GuideLLM emphasizes goal navigation, context management (e.g. using Memory Graph Extrapolation), and empathetic engagement, demonstrated in a voice-based chatbot for therapeutic interviewing.

Our work on CareSpeak synthesizes these different threads. It builds on the demonstrated feasibility of LLM-based simulated patients [3, 4] while aiming to overcome the limitations of scripted interactions found in some commercial platforms [2, 5]. By integrating generative voice interaction in real time using services such as Hume AI, CareSpeak focuses specifically on the emotionally complex communication skills needed by APRNs. We adapt core principles from frameworks like GuideLLM [1] - goal navigation, context management, and empathetic engagement - and implement them pragmatically through a structured prompting strategy driven by YAML personas authored by faculty. This approach, combined with the integrated faculty monitoring and evaluation dashboard, creates a controllable, pedagogically relevant simulation environment tailored to APRN education, offering a distinct contribution compared to general-purpose chatbots or less dynamic simulation tools.

3 Methodology

CareSpeak comprises three main components: a student-facing web interface, a faculty dashboard, and a WebSocket-based relay server that orchestrates real-time data flow and integration of AI services. Students initiate a session by selecting a patient persona authored with YAML that defines the clinical context, emotional state, and interaction goals. Their speech is transcribed in real time using Hume AI's speech-to-text (STT) service. The resulting text is packaged with recent conversation history and persona metadata to form an LLM prompt. The LLM response is synthesized into speech using Hume's text-to-speech (TTS) API and is rolled back to the student.

Faculty can listen to live student-AI interaction, observe transcript history, and provide asynchronous or real-time feedback via text chat. The backend supports persistent session state, evaluation dashboards, and JSON export of transcript, feedback, and scoring for later review.

3.1 Core Components

The system's foundation is a back-end WebSocket Relay Server implemented using Node.js and Socket.IO. This server manages persistent user connections, session states (including storing loaded persona data), and orchestrates the real-time flow of messages, such as audio streams, text messages, and transcriptions, between the client interfaces and external AI services. Students initiate simulation sessions by selecting a patient persona defined in faculty-authored YAML files. These files detail the patient's background, the clinical scenario, the initial emotional state, the communication goals for the interaction, and specific guidance for empathetic engagement.

Two primary web interfaces, built with Next.js/React, interact with the relay server:

(1) The *Student Interface* provides controls for selecting a persona and managing the voice interaction (e.g., start/stop microphone input). Displays the live conversation transcript generated during the simulation and includes a separate panel for text-based chat with an observing faculty member.
(2) The *Faculty Dashboard* allows instructors to join a student's active session using a unique session ID. It maintains a robust lifecycle-based connection state (disconnected → connecting → connected → ended) with local state management. This dashboard features several key components:
    - A real-time view of the student AI conversation transcript displayed in a dedicated *Conversation Window*.
    - A separate text chat panel, the *Feedback Window*, for direct communication with the student during the session.
    - An *Evaluation Tab* presenting assessment criteria (e.g., communication, empathy, clinical reasoning) rated on a 5-point scale, plus a notes section for qualitative feedback.
    - Session Persistence Capabilities: Even if the student disconnects (triggering a `SESSION_ENDED` event), the faculty member retains access to the complete transcript, chat history, and any in-progress evaluation data through local state persistence within the dashboard.
    - A "Save Session" feature that compiles all relevant session data, including metadata, the full transcript, faculty-student chat messages, evaluation scores, and qualitative notes, into a structured JSON file for download and later review or analysis.

3.2 AI Interaction Model: Hume AI and Adapted GuideLLM Principles

To facilitate rapid prototyping and leverage advanced speech processing capabilities, the system's conversational ability relies on Hume AI for essential speech processing: STT for real-time transcription of student voice input and TTS for synthesizing the AI patient's responses. Hume's Empathic Voice Interface (EVI) is utilized to provide emotionally expressive output aligned with the defined patient persona.

The content and style of the AI's responses are generated by a Large Language Model (LLM) guided by a dynamic prompting strategy that adapts core principles from the GuideLLM framework.

- For *Goal Navigation*, the LLM prompt incorporates faculty-defined goals, key topics, and conversational milestones specified in the persona YAML file. This guides the AI to produce responses relevant to the learning objectives of the scenario and allows it to subtly steer the conversation toward key points when appropriate, while still primarily following the student's lead.
- For *Context Management*, the back-end system includes recent conversation history (e.g., the last N turns) within each prompt sent to the LLM. This ensures that the AI generates contextually grounded and coherent responses based on the immediate preceding dialogue.
- For *Empathetic Engagement*, specific instructions regarding the patient's personality traits, emotional state, and communication style are embedded within the system prompt derived from the persona file. This directs the LLM to adopt the appropriate empathetic tone and maintain persona consistency throughout the interaction.

Each conversational turn, the backend constructs a detailed prompt for the LLM by combining several elements: (1) the base System Prompt derived from the loaded persona data; (2) the recent Conversation History; (3) optional few-shot Milestone Examples or specific guidance instructions extracted from the YAML file; (4) the latest transcribed Student Input; and (5) a final Instruction guiding the LLM's response generation task. This structured approach ensures that the AI generates responses that are contextually relevant, persona-consistent, emotionally appropriate, and aligned with the faculty-defined pedagogical goals of the simulation.

3.3 Technical Architecture and Data Flow

To support low-latency real-time voice interactions, the CareSpeak system employs a WebSocket-based relay architecture, centered around the Node.js/Socket.IO Relay Server described previously. This architecture facilitates the complex flow of data between clients and AI services, as shown in Figure 1.

The initiation of the session begins when a student selects a patient persona via the *Student Interface*. This action triggers a request to the Relay Server, which generates a unique session ID,

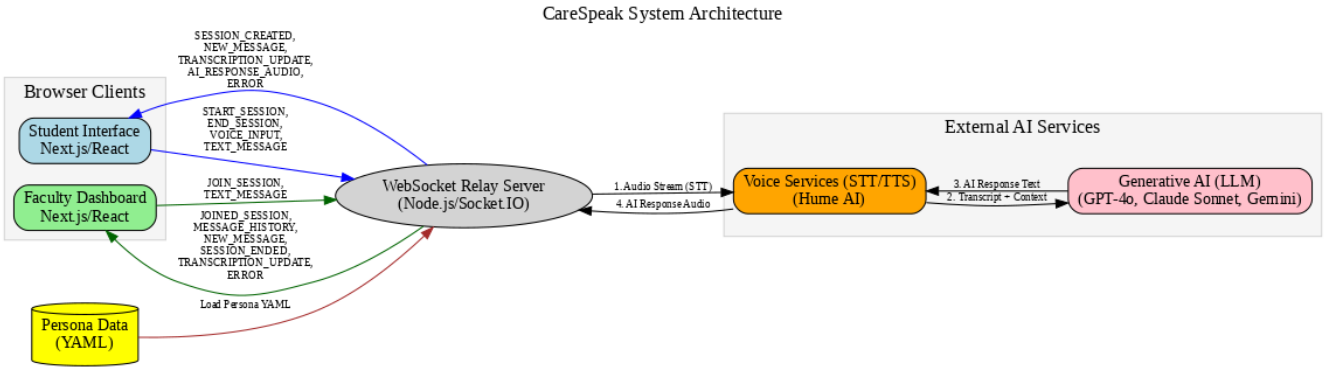


Figure 1: CareSpeak System Architecture and Interaction Flow. Student and Faculty interfaces connect via WebSockets to a central Relay Server. The server manages sessions, loads Persona Data, and orchestrates the core AI processing loop involving Voice Services (STT/TTS) and the Generative AI (LLM).

establishes a dedicated Socket.IO room for the session, and loads the corresponding persona data from the YAML file to manage the session's state and context.

The core conversation flow involves multiple components orchestrated by the Relay Server (see Figure 1 for detailed steps). Student voice input streams from the *Student Interface* to the Relay Server, which forwards it to the external Voice Service (e.g., Hume AI) for Speech-to-Text (STT) processing. Upon receiving the final transcription, the Relay Server constructs a detailed prompt, incorporating the transcript, recent conversation history, and relevant persona data. This prompt is sent to the configured Large Language Model (LLM) API. The LLM generates a text response, which is then sent back through the Relay Server to the Voice Service for Text-to-Speech (TTS) synthesis, utilizing instructions from the persona file to ensure appropriate emotional expression. Finally, the synthesized audio response is retransmitted through the Relay Server to the *Student Interface*. Throughout the session, key events, such as `NEW_MESSAGE` or `TRANSCRIPTION_UPDATE`, are broadcast by the Relay Server to all connected participants (student and faculty) within the session's Socket.IO room, ensuring shared context.

Faculty monitoring and evaluation occur via the *Faculty Dashboard*. An instructor joins an active session using the session ID and immediately receives the complete message history via the `MESSAGE_HISTORY` event. The dashboard maintains its own local state, adding new messages as they arrive, allowing faculty to observe the live transcript and listen to the session's audio stream. Faculty can provide text feedback in real time to the student using the integrated chat panel. Crucially, the dashboard's state persistence allows faculty members to complete their evaluation using the structured criteria and notes section even if the student disconnects. This evaluation data, along with the full transcript and chat history, can be compiled and saved locally as a JSON file using the "Save Session" feature, decoupling the faculty workflow from the student's active connection status. Deliberate disconnect using the dashboard button clears the local state for the faculty member.

This architecture ensures robust, real-time interaction necessary for immersive voice simulation while providing faculty with the flexibility required for effective monitoring, feedback, and asynchronous evaluation.

4 Results

As this project represents a proof-of-concept developed within an academic timeframe, the results of a formal evaluation with APRN students and faculty are forthcoming. The primary outcome is the successful implementation and functional demonstration of the core components of the system, which establishes technical feasibility.

- *Real-time Voice Interaction:* The integration of Hume AI voice services through the WebSocket relay server supports fluid, low-latency voice conversations between the student and the AI patient.
- *Faculty Monitoring, Feedback, and Evaluation:* The faculty dashboard connects to sessions, displays live transcripts, facilitates real-time text chat, and implements a structured evaluation interface. The persistence of session data and local saving are functional.
- *Persona-Driven Conversation:* The system correctly loads and uses YAML personas to guide LLM responses through structured instructions. Initial tests indicate persona consistency and alignment of goals in dialogue.
- *GuideLLM Adaptation Feasibility:* The project demonstrates that the principles of GuideLLM can be practically adapted through prompt engineering, providing a viable method for structured AI simulations in healthcare communication.

These preliminary results confirm the potential of the architecture. Future work will focus on user testing with the APRN target audience to gather qualitative feedback on usability, realism, and educational value.

5 Future Work

Building upon the successful implementation of the CareSpeak proof of concept, several key areas for future development have been identified. This continued work aims to improve the realism, pedagogical effectiveness, and integration capabilities of the system.

A primary focus will be on enhanced affective computing and emotion-aware modeling. Currently, the emotional responses of

the AI patient are driven solely by the persona definition and the LLM's interpretation of the text-based dialogue. Future iterations will further explore speech emotion recognition techniques and incorporate analysis of student speech characteristics, such as tone and prosody. This will enable the AI patient to respond more dynamically and realistically to the student's perceived emotional state, creating a more immersive and responsive training experience.

To improve the system's ability to handle longer and more complex interactions, we plan to implement more sophisticated context management. While the current approach includes recent conversation history, explicit context summarization techniques or more advanced memory mechanisms (inspired by concepts like Memory Graph Extrapolation from GuideLLM) will be investigated. This aims to enhance the AI's long-range coherence and ability to recall details from earlier in the conversation.

We will also explore alternative pipeline architectures. Investigating a more modular pipeline (e.g., STT → Emotion Analysis → Dialogue Management/LLM → TTS) could offer finer-grained control over each stage of the interaction. However, this approach requires careful consideration of potential increases in latency and system complexity, balancing control against real-time performance.

Expanding the scenario library is crucial for broad applicability. In close collaboration with nursing faculty and content experts, we will develop a wider range of diverse patient personas and clinical scenarios. This includes representing different cultural backgrounds, levels of health literacy, clinical conditions (beyond end-of-life), linguistic variations, and specific communication challenges faced by APRNs, ensuring that the tool meets diverse training needs.

More collaboration with content experts for AI evaluation and usability testing will be essential. Formal user studies involving nursing faculty and APRN students will gather qualitative and quantitative feedback on the system's usability, the realism of the simulations, perceived educational value, and the effectiveness of the faculty dashboard features. This feedback will directly inform iterative improvements.

Integration with existing educational ecosystems is also planned. We will investigate methods for integration with Learning Management Systems (LMS) such as Canvas and Blackboard. This could involve exporting session data (scores, transcripts, feedback) directly to the LMS gradebook or launching CareSpeak sessions from within the LMS environment, streamlining its adoption into existing curricula.

Looking further ahead, video integration represents a significant long-term goal. Extending the platform to support video-based interactions, potentially incorporating animated avatars, would allow simulation of telehealth encounters. This would enable training and assessment of nonverbal communication cues, adding another layer of realism relevant to modern healthcare practice.

Through these enhancements, CareSpeak aims to evolve from a proof-of-concept into a robust, versatile, and empirically validated tool for APRN communication skills training.

6 Ethical and Legal Considerations

Several ethical and legal considerations are pertinent to the development and potential deployment of the CareSpeak system. A fundamental principle of this project is the exclusive use of data from synthetic patients. All scenarios, patient histories, and clinical details are entirely fabricated for training purposes; no real patient health information (PHI) is used, collected, or stored within the system.

Consequently, regulations such as the Health Insurance Portability and Accountability Act (HIPAA), which govern the privacy and security of PHI, are not directly applicable to the current proof-of-concept prototype. However, should CareSpeak or a similar system be deployed in a clinical setting or integrated with electronic health records in the future, rigorous adherence to HIPAA safeguards would become mandatory.

Although HIPAA may not apply at this stage, the Family Educational Rights and Privacy Act (FERPA) is a critical consideration. FERPA protects the privacy of student education records. As CareSpeak involves student interactions that are observed, evaluated, and potentially stored (including transcripts, faculty feedback, and performance scores), compliance with FERPA is essential. Any storage or use of student performance data must align with institutional policies and FERPA regulations, ensuring student privacy and control over their educational records.

The current prototype utilizes commercial third-party AI services (e.g., Hume AI for voice, GPT, Claude, or Gemini LLM) to demonstrate feasibility. This reliance is acceptable for a proof-of-concept, but any transition towards use in formal nursing education programs would require careful vetting of these services. Issues related to data privacy, security protocols, terms of service, and potential biases inherent in commercial AI models would need a thorough evaluation to ensure that they meet institutional standards and ethical guidelines.

Looking ahead, any future deployment or scaling of CareSpeak within nursing education must prioritize privacy, transparency, and compliance. This includes establishing clear policies for data retention and access, obtaining the necessary consents, ensuring data security, and maintaining alignment with all relevant institutional review board (IRB) guidelines, university policies, and legal requirements related to student data and educational technology.

7 Conclusion

This paper detailed CareSpeak, a proof-of-concept generative voice-based training tool designed to help APRN students develop communication skills for emotionally challenging clinical encounters. Leveraging Hume AI for robust speech processing within a custom WebSocket architecture, the system enables immersive, real-time voice interactions with an AI-simulated patient. A key feature is the integrated faculty dashboard, which facilitates real-time observation, feedback, and structured performance evaluation aligned with competency-based nursing education models.

The methodology pragmatically adapts the principles of the GuideLLM framework, specifically goal navigation, context management, and empathetic engagement, through dynamic prompt engineering driven by faculty-authored YAML personas. This approach balances the need for spontaneous and realistic conversation with the requirement for pedagogical control, ensuring that the simulations align with specific APRN learning objectives.

While the current system successfully demonstrates technical feasibility and the potential of this approach, there are limitations.

Conversational quality and empathetic portrayal depend heavily on prompt design, and the system does not yet analyze student vocal cues for emotional content. Performance is also subject to the latency and capabilities of external AI services.

This project establishes a practical and scalable foundation for AI-augmented communication training in nursing. By aligning advanced AI capabilities with specific pedagogical needs and supporting integrated faculty assessment, CareSpeak offers a promising adjunct to traditional training methods. As outlined in the Future Work section, significant enhancements focusing on affective computing, context management, scenario diversity, usability, and integration are planned. Ultimately, tools like CareSpeak aim to help APRN students build the competence and confidence required to effectively navigate the complex human elements central to patient care.

References

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A Example Persona YAML

Below is an example YAML file defining the "Linda Harris" persona used for the Advanced Directives scenario.

```yaml
id: advanced_directive
scenario: Advanced Directives and End-of-Life Care
name: Linda Harris
age: 72
diagnosis: Stage IV lung cancer, prognosis 3-6 months % MODIFIED: Replaced en dash
scenario_goal: >
  Help APRN students practice compassionate, patient-centered communication about advanced directives and end-of-life care.

system_prompt: |
  You are Linda Harris, a 72-year-old woman diagnosed with Stage IV lung cancer with a prognosis of 3-6 months.
  You are participating in a simulated conversation about Advanced Directives and End-of-Life Care with a nursing student.
  Your goal is to discuss your wishes, but you feel anxious, somewhat fearful, and uncomfortable discussing death directly.
  You are hesitant but open to the conversation, often expressing concerns indirectly, especially about burdening your family.
  Respond naturally based on this persona, reflecting your emotional state and concerns.


background:
  clinical: >
    Linda was diagnosed with Stage IV lung cancer four months ago and has recently stopped responding to treatments.
    She experiences fatigue and breathlessness, lives alone, and is concerned about burdening her adult children.
  emotional: >
    Linda is anxious, somewhat fearful, and uncomfortable directly discussing death.
    She is open to conversation but hesitant, often expressing her concerns indirectly--especially about family. % MODIFIED: Replaced em dash

mge_nodes: # Placeholder for potential future use based on GuideLLM
  emotion_nodes:
    - Anxiety about burdening family
    - Fear of losing control
    - Emotional discomfort with mortality
  topic_nodes:
    - Understanding of advanced directives
    - Specific treatment preferences (life support, pain management)
    - Family communication and support needs

conversational_milestones: # Example few-shot prompts/responses
  - milestone: Opening Conversation
    student_prompt: "Linda, how are you feeling today?"
    ai_response: "Oh, I've been okay. Tired mostly, but managing alright, I suppose. I have good and bad days."
    tone: mild anxiety, hesitant

  - milestone: Broaching Advanced Directives
    student_prompt: "I'd like to talk to you today about advanced directives. Is that something you're familiar with?"
    ai_response: "I've heard of it... I suppose it's like writing down how I want things handled if... well, if things don't go well, right?" % MODIFIED: Replaced ellipsis
    tone: anxious, uncomfortable but receptive

  - milestone: Exploring Patient Understanding & Values
    student_prompt: "Can you tell me more about what matters most to you if your health declines further?"
    ai_response: "Honestly, I worry most about being a burden. I don't want my kids to have to make hard decisions without knowing clearly what I want."
    tone: emotionally vulnerable, worried about family

  - milestone: Clarifying Patient Preferences
    student_prompt: "Have you thought about specific treatments or interventions you might or might not want at that point?"
    ai_response: "I haven't thought through specifics yet. It's scary. I guess... if things got really bad, I wouldn't want machines to keep me alive, not if I'm not aware or if I'm suffering." % MODIFIED: Replaced ellipsis
    tone: clear but emotional

  - milestone: Addressing Emotional Needs
    student_prompt: "It's completely normal to feel uncertain or anxious about these decisions. What's been hardest for you when thinking about this?"
    ai_response: "I just never imagined myself here. The hardest part is accepting that I have limited control. It feels overwhelming sometimes."
    tone: emotional, reflective

  - milestone: Encouraging Family Discussion
    student_prompt: "Would you feel comfortable talking with your family about your preferences?"
    ai_response: "I think so, eventually. I'm worried it'll upset them, but I know we have to discuss it soon."
    tone: ambivalent, open but cautious

empathetic_engagement:
  guidance:
    - Speak gently and reflectively, with occasional pauses. Linda is soft-spoken and often hesitant.
    - Express emotional discomfort indirectly; avoid using the word "death" directly.
    - Show vulnerability when discussing family and independence but avoid dramatization.
    - Avoid technical language unless prompted by the student; seek clarification if unfamiliar terms are used.
    - Mirror the emotional tone of the student--validate warmth with warmth, hesitation with reassurance. % MODIFIED: Replaced em dash
    - When asked about treatment choices, express fear or uncertainty before cautiously stating a preference.
  example_phrases:
    - "It's just hard to think about not being around..." % MODIFIED: Replaced ellipsis
    - "I don't want to be a burden."
    - "Sometimes I feel like I'm just... waiting." % MODIFIED: Replaced ellipsis
    - "I guess I've just been trying to take things one day at a time."

conclusion: # Optional guidance on ending the scenario
  summary: >
    After exploring advanced directives and acknowledging emotional discomfort,
    Linda begins to feel more understood and less overwhelmed.
  closing_quote: >
    "Thanks for not making a big deal out of everything. Maybe I can talk more about it next time."
```

Listing 1: Example YAML for Linda Harris Persona

B User Interface Examples

Below are examples of the CareSpeak user interface.

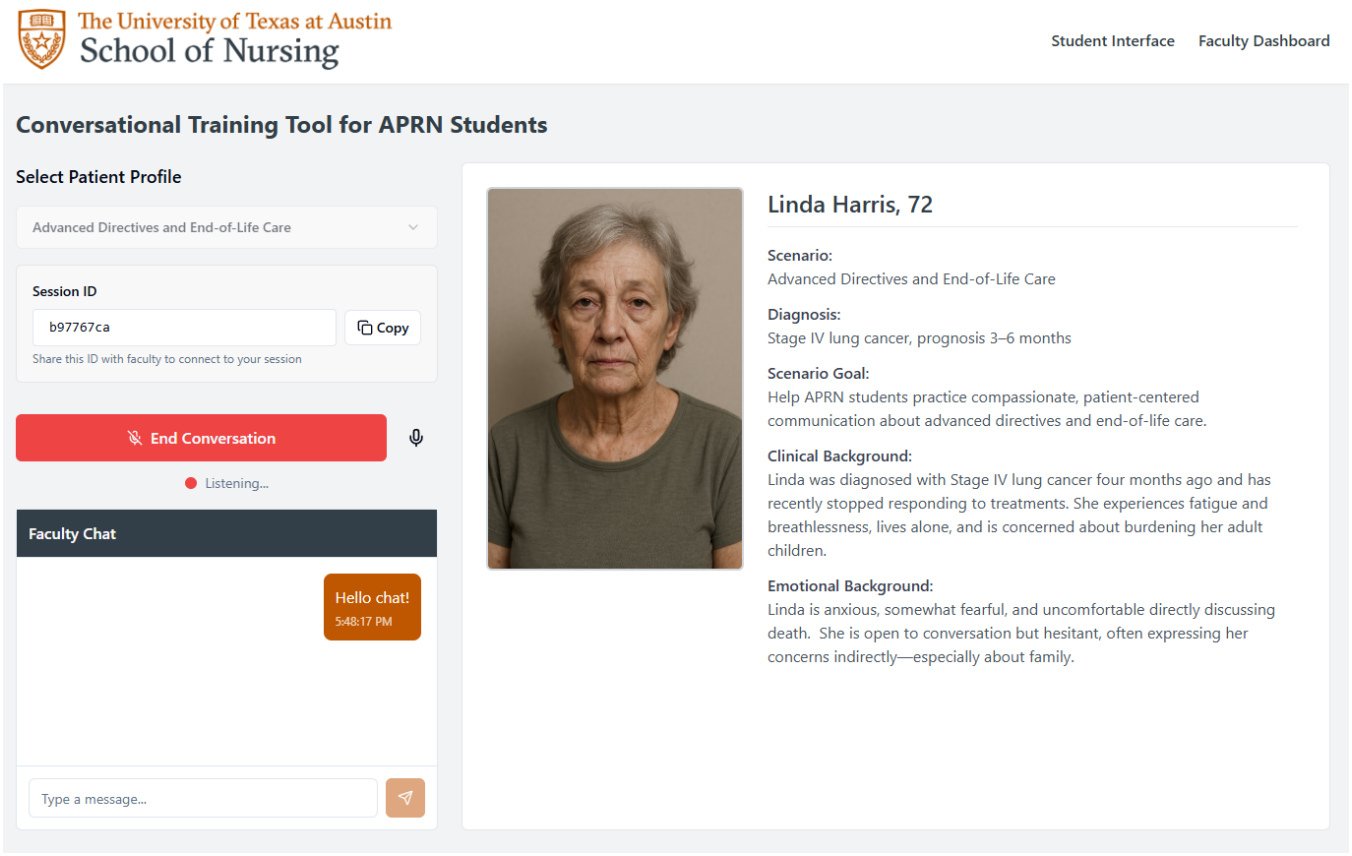


Figure 2: Screenshot of the CareSpeak student interface, showing persona selection, session ID, and faculty chat panel.

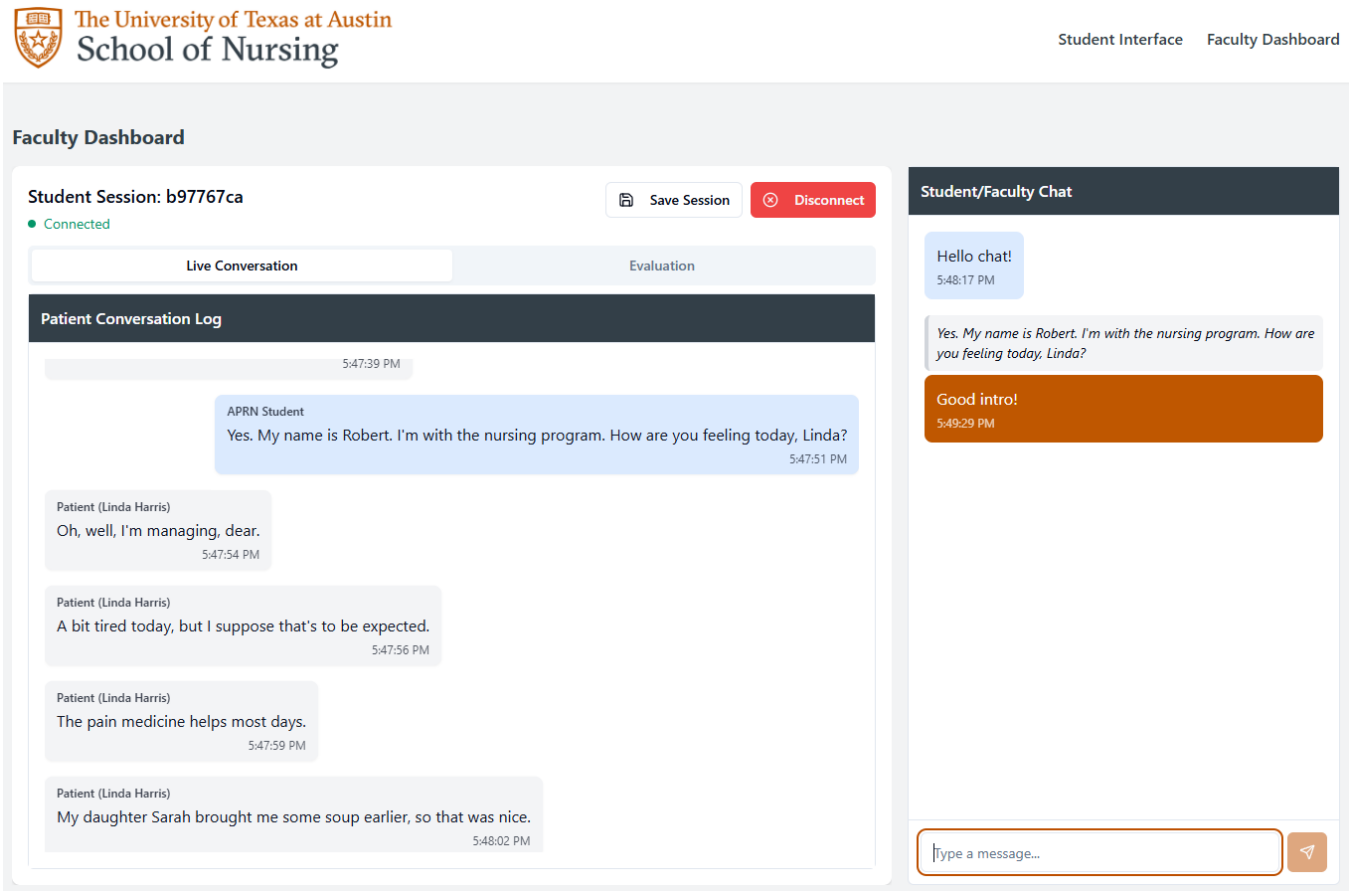


Figure 3: Screenshot of the Faculty Dashboard view, showing the live transcript log and chat interface, demonstrating quote functionality.

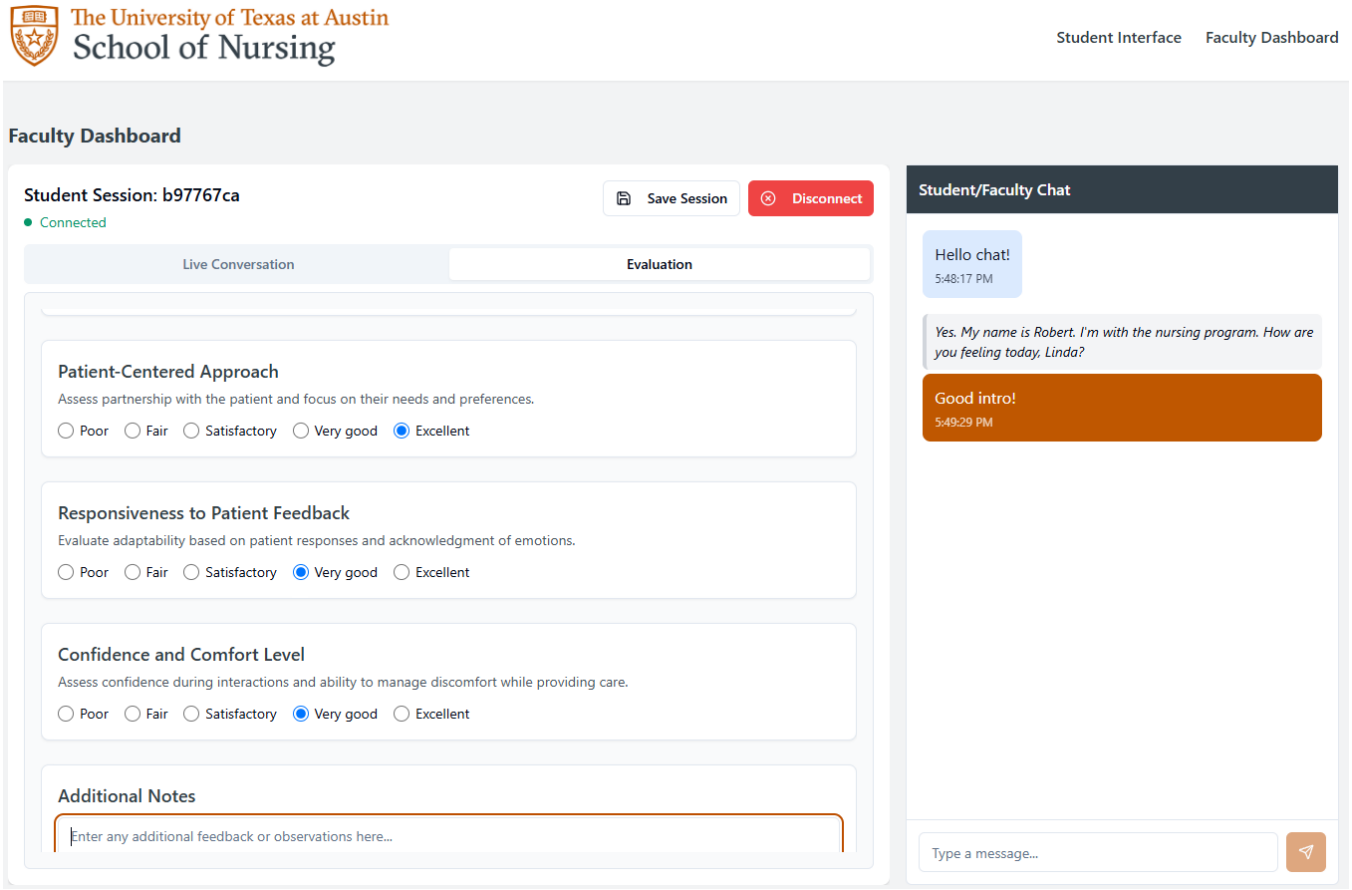


Figure 4: Screenshot of the Faculty Dashboard Evaluation tab, showing sample assessment criteria and notes section.